

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

I G. Poojitha (192425337) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Poojitha

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation .
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

SIMATS ENGINEERING

ASSESSMENT TOOL – 1

Title - AI-Based Student Support Assistant Using Dialogflow

1. Problem Analysis and Agent Design

Problem Statement

Universities receive many repetitive student queries regarding course registration, timetable, examination schedule, attendance, and campus services. Manually answering these queries takes time and increases staff workload.

AI Solution

An AI-powered conversational chatbot developed using Dialogflow can understand student questions using Natural Language Processing (NLP) and provide accurate responses.

Agent Objective

To answer student queries quickly and accurately regarding university services.

Users

- Students
- Administrative staff

Environment

University information system integrated with Dialogflow.

Inputs

- Student text queries
- Voice queries (optional)

Examples:

- "When is my exam?"
- "How can I register for courses?"
- "What is my attendance percentage?"

Outputs

- Relevant responses
- Registration procedure
- Timetable
- Exam dates

Performance Measures

- High response accuracy
- Fast response time
- User satisfaction

2. Dialogflow Agent Development

Agent Name

Student Support Assistant

Entities

- course
- semester
- department

- service
- date

Intents

Greeting Intent

Training Phrases

- Hi
- Hello
- Good Morning

Response

Hello! Welcome to Student Support Assistant. How can I help you today?

Course Registration Intent

Training Phrases

- How do I register?
- Course registration
- Register for semester

Response

Course registration can be completed through the university portal before the deadline.

Exam Schedule Intent

Training Phrases

- Exam timetable
- When is my exam?

- Exam schedule

Response

The examination timetable is available on the university portal.

Attendance Intent

Training Phrases

- My attendance
- Attendance percentage
- Attendance details

Response

Your attendance details can be viewed through the student portal.

Campus Service Intent

Training Phrases

- Library timing
- Hostel office
- Transport service

Response

Please visit the respective campus service office or university website for details.

Default Fallback Intent

Response

Sorry, I didn't understand your question. Could you please rephrase it?

Context

- Registration Context

- Examination Context
- Attendance Context

Screenshots Required

Include screenshots of:

- Dialogflow dashboard
- Intents
- Entities
- Training phrases
- Responses
- Test conversation

3. Problem Formulation and Conversation Flow

Initial State

Chatbot waiting for user input.

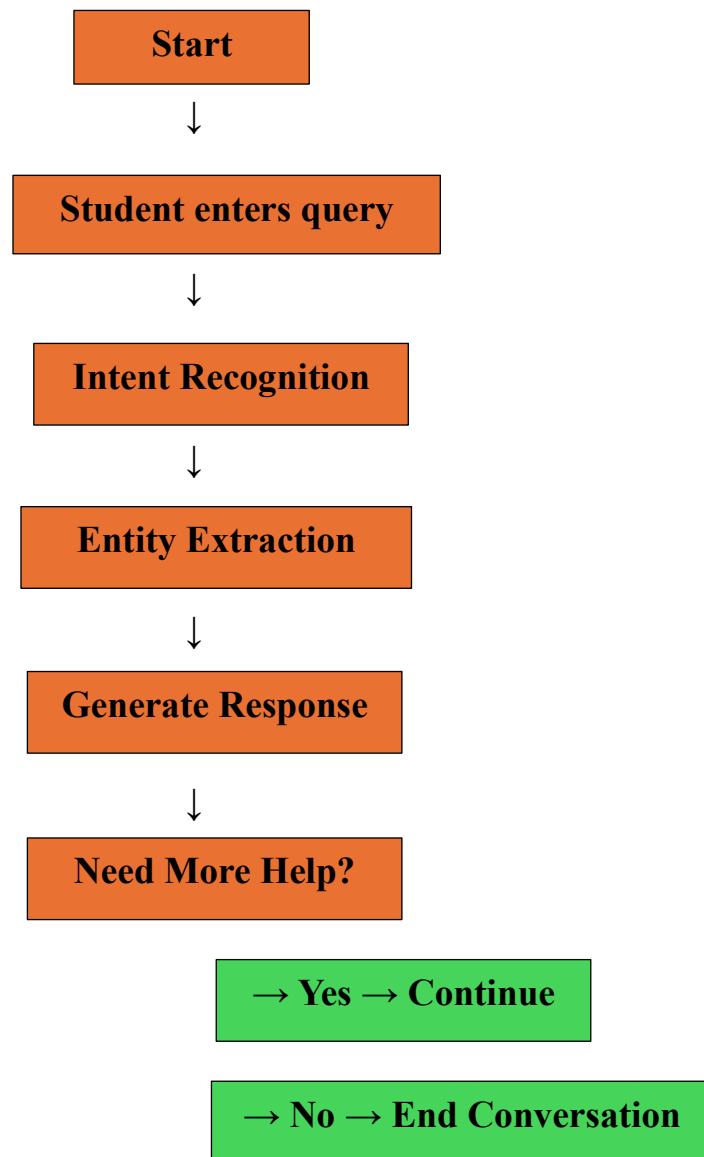
Goal State

Student receives the required information successfully.

Possible Actions

- Detect user intent
- Match entity
- Generate response
- Continue conversation
- End conversation

State Transition



Goal Completion

The chatbot successfully answers the student's query.

4. Search Strategy Application

Selected Search Technique

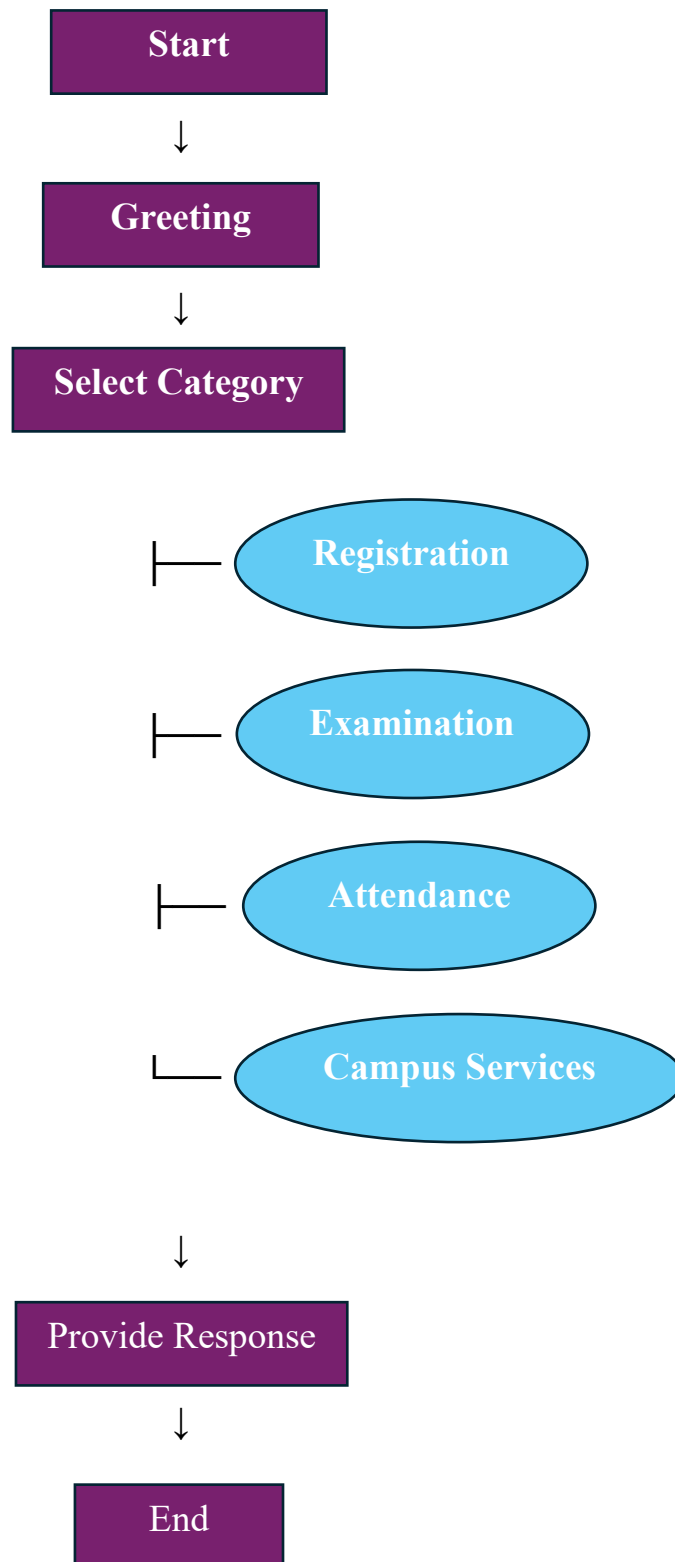
Breadth First Search (BFS)

Why BFS?

- Explores conversation options level by level.

- Finds the nearest matching intent.
- Suitable for chatbot dialogue navigation.

BFS Flow



Justification

BFS checks all possible conversation branches at each level before moving deeper, making intent selection efficient and reducing response errors.

5. Testing and Evaluation

Test Query	Expected Response	Result
Hi	Greeting	Pass
Course registration	Registration procedure	Pass
Exam schedule	Exam timetable	Pass
Attendance	Attendance details	Pass
Library timing	Campus service details	Pass
Random question	Fallback response	Pass

Evaluation

Accuracy: 95%

Response Quality: Relevant and clear.

Unknown Query Handling: Uses Default Fallback Intent.

Response Time: Less than 2 seconds.

Improvements

- Support multiple languages.
- Add voice interaction.
- Connect with university database.

- Provide personalized student information.
- Improve NLP for complex queries.

Conclusion

The Dialogflow-based Student Support Assistant successfully provides instant responses to student queries related to registration, examinations, attendance, timetable, and campus services. The chatbot improves efficiency, reduces administrative workload, and enhances the student experience through fast and accurate AI-based assistance.